

Zhaode Wang

AI Inference Engine Expert · On-Device LLM · High-Performance Computing

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Education

Institute of Computing Technology, Chinese Academy of Sciences	M.S. in Computer Architecture	2017 - 2020
State Key Laboratory of Computer Architecture		
Shandong University	B.S. in Computer Science and Technology	2013 - 2017

Experience

Alibaba · Taotian Group	Senior Technical Expert · MNN Tech Lead	2020 - Present
MNN core architecture: Lead MNN architecture and heterogeneous performance optimization across ARM CPUs, Metal/OpenCL/Vulkan GPUs, and Hexagon/CoreML/NNAPI accelerators, covering operator kernels, memory planning, graph optimization and fusion, and model conversion. On-device AI platform: Built an embedded Python runtime and end-to-end deployment stack, enabling dozens of algorithms for Pailitao visual search, Taobao Live, security, and feed; supported Quark, Qwen, DingTalk, Youku, Taobao Instant Commerce, and Amap; grew MNN to 15.8K+ GitHub stars. MNN-LLM & production: Initiated and currently lead MNN-LLM for 100+ language and multimodal models, spanning 2/3/4/8-bit and mixed-precision quantization, KV-cache and attention optimization, and speculative decoding; deployed on-device LLMs in Taobao and Xianyu search and recommendation at tens-of-millions-user scale. On-device SLM: Led a team to train a 0.3B SLM from scratch for on-device applications, covering data collection and quality assessment, pre-training, SFT, RL, and end-to-end model evaluation.		
Cambricon Technologies	Compiler Engineer Intern	2018 - 2020
Developed compiler and linker components for Cambricon's BANG toolchain, focusing on heterogeneous linking; subsequently optimized Caffe operators for MLU accelerators.		
Megvii	Algorithm Engineer Intern	2019
Optimized traffic video understanding algorithms and productionized them in C++.		
Microsoft Asia Engineering Academy	Software Engineer Intern	2016
Evaluated LevelDB, Presto, and Hadoop for advertising data storage and analytics workloads.		

Selected Publications

Core Contributions		
Efficient LLM Inference on ARM via Hardware-Aware Operator Co-Design and Heuristic Mixed-Precision Search	ISCAS	2026
RecGPT-Mobile: On-Device Large Language Models for User Intent Understanding in Taobao Feed Recommendation	SIGIR	2026
MNN-LLM: A Generic Inference Engine for Fast Large Language Model Deployment on Mobile Devices	MM Asia	2024
Walle: An End-to-End, General-Purpose, and Large-Scale Production System for Device-Cloud Collaborative Machine Learning	OSDI	2022
Collaborative Research		
LISA: Locality-Informed Speculative Decoding for Accelerating Autoregressive Image Generation	ECCV	2026
ARC-Decode: Accelerated Decoding with Risk-Bounded Acceptance	ICML	2026
Prism-MoE: Efficient Dense-to-MoE Conversion for Visual Autoregressive Generation	ICML	2026
AccKV: Towards Efficient Audio-Video LLMs Inference via Adaptive-Focusing and Cross-Calibration KV Cache Optimization	AAAI	2026
VertiKV: Vertical-Integrity KV Cache Compression for Efficient Multimodal Long-Context Inference	ACM MM	2026
MadaKV: Adaptive Modality-Perception KV Cache Eviction for Efficient Multimodal Long-Context Inference	ACL	2025

Patents & Applications

BANG-Linker: Heterogeneous Linking Method, Apparatus, and Related Products	Cambricon · Filed	2019
BANG Simulator: Programming Debugging Method, Apparatus, and Related Products		
Solution for Deploying Large Language Models on Mobile Devices	Alibaba · Filed	2024
Saliency-Aware Blockwise Quantization via Floating-Point Zero-Point Degrees of Freedom	Alibaba · Pending	2026
Mixed-Precision Allocation for Low-Bit LLMs Based on Action-Conditioned Operator Error		

Honors & Competitions

IEEE AICAS Grand Challenge: LLM Software and Hardware System Co-optimization	First Prize	2024, 2025
NPU Competitions: Tecorigin Operator Development & SOPHGO TPU Programming	First Prize	2022, 2024

Skills

Languages	C, C++, Python, Assembly, CUDA, OpenCL, Vulkan, Metal, BANG, Java, Objective-C
Expertise	Compilers, AI inference engines, CPU/GPU/NPU operator optimization, model conversion, quantization, large language models